

WEICHEN HUANG

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EDUCATION

Valley Christian Schools

San Jose, CA · Expected 2027

GPA: **3.87** Unweighted / **4.63** Weighted · SAT: **1490**

Relevant Coursework: AP Statistics, AP Physics, AP Calculus BC, Multi-Variable Calculus, AI/ML Course, Blue Ocean Student Entrepreneurs Mini-Course

Google AI Professional Certificate (Coursera) — 7-course program demonstrating fluency in real-world data analysis, app building, and complex workflow automation

Blue Ocean Student Entrepreneurs Mini-Course — Specialized program on value innovation and frameworks for identifying uncontested market spaces and designing disruptive business strategies

COMMUNITY SERVICE

Sound Engineer & Technical Coordinator — Lincoln Glen Church

San Jose, CA · 2 Years

- Managed sound engineering and AV logistics for weekly Sunday services, ensuring every attendee experienced a **high-quality, distraction-free service**
- Collaborated with the worship team and stage crew to troubleshoot live technical issues in real time, holding to the standard that **technology failing people in a moment that matters is never acceptable**

LEADERSHIP

Senior Lead of Operations — Walley's School Store, BEI Dept.

Incoming 2026–2027

- Promoted to direct strategic initiatives and oversee a **5-person operations team** for a **gold-certified** student-run enterprise during the upcoming academic year
- Planning a **physical token-card program** to convert parent pop-up traffic into concession revenue, designing category pricing, Square configuration, redemption controls, and a four-semester revenue forecast from a full year of transaction data (**9,300+ transactions**)
- Organizing an **expanded pop-up event calendar** across both semesters for 2026-27, including a new spring slate the store has never run, after analysis showed event days drive **41% of annual revenue**

Lead of Operations — Walley's School Store, BEI Dept.

1 Year

- Architected and deployed a **full-stack TypeScript and Rust analytics platform** used daily by a 9-person team to process, visualize, and audit **8,654 transactions** totaling **\$45,922 collected** (\$42,674 net sales)
- Automated error detection in Square POS data, **reducing inventory discrepancies by 42%** and surfacing errors previously undetected across thousands of transactions
- Engineered and launched a campus-wide mobile vending program that expanded store reach to multiple campus zones, driving a **15% increase in weekly revenue**
- Analyzed POS sales velocity and seasonal demand to identify underserved product categories, curating new inventory items that **expanded the store's catalog and drove incremental revenue growth**
- Transitioned the store to a dynamic 2-week purchasing cycle driven by POS sales velocity, optimizing cash flow and **systematically preventing stockouts**
- Directed bi-weekly inventory audits, reconciling physical stock against digital records to minimize shrinkage and safeguard **\$45,922** in total collected revenue across the year

Operations Officer — DECA

1 Yr Officer, 2 Yrs Member · ICDC Qualifier

- Built the **event and judge scheduler** the chapter runs its mock conferences on, automatically assigning students to events, matching judges by expertise, and allocating rooms for a **200+ member** chapter
- Built the chapter's **registration-form checker** (Python + OCR) that verifies the **~600 signed PDFs** collected each year, replacing **10+ hours** of manual review each cycle by flagging exactly which forms and fields need resubmission
- Improved the chapter's matching tools — rewrote the mentor-pod sorter and repaired the roommate-assignment pipeline — so mentees and conference rooms are assigned automatically instead of by hand

Operations Officer — YHCCA

Chinese Arts & Culture Nonprofit · 2026

- Rebuilt the organization's **website** so members and the public can find event dates, program details, and how to get involved without having to email first
- Built **budget spreadsheets** for two upcoming events, an art gallery and an art contest, tracking line-item costs against committed spend
- Sourcing **venues** for the gallery, comparing cost and capacity against the show's actual needs

RESEARCH

AI Researcher: Algorverse

June 12 — August 2026

- Selected for a competitive AI research program; developing **Conformal Set Inversion**, a method for producing **distribution-free uncertainty guarantees** on hidden parameters that are never directly observed at decision time
- Core idea: calibrate a conformal prediction band in observable space, then invert it through the forward model to bound the latent state, **removing the need for a trained posterior estimator** and staying robust when the model is misspecified
- Differentiating the approach from prior work (CP4SBI) by **explicitly budgeting for surrogate-model discrepancy** and using Jacobian singular-value structure to select held-out coordinates as both a design and a diagnostic tool
- Co-authored the paper and **submitted it to three NeurIPS 2026 workshops (under review)**; ran experiments on a simulation-based inference testbed with a research partner and faculty mentors

WORK EXPERIENCE

Assistant Store Manager — Walley's School Store

2 Years

- Oversaw daily transactions and inventory for a student-run retail/food enterprise generating over **\$1,000 in weekly revenue** across **8,654 annual transactions**
- Leveraged Square POS data to monitor sales velocity, identify peak demand windows, and inform restocking decisions, **reducing stockouts during high-traffic periods**
- Fostered a collaborative shift environment by **training incoming volunteers**, coordinating team schedules, and facilitating smooth handoffs between operational pods

Business & AI Summer Camp Instructor

June 1 — July 17, 2026

- Led an in-person summer program for **40 students**, teaching them to run every part of Walley's School Store from the ground up: **opening procedures, register operations, inventory management**, and team-based workflows
- Guided students through the merchandising side of retail, using sales data and real demand to decide **which products were worth stocking** rather than guessing
- Ran hands-on production sessions where students **designed and heat-pressed their own merchandise**, taking a product from concept to a finished item ready for the shelf
- Culminated the program in a mock grand opening, the **Walley's Summer Camp Open House**, where student teams operated the store live for real customers
- **Four student-created products** performed well enough over the summer that they are now a **permanent part of Walley's official menu**

Coding Tutor — Independent

3 Months

Curriculum · slide deck

- Shared an open invitation in community **WeChat groups** to offer free coding lessons, which connected me with **20 students** who had no prior programming background
- Held weekly 2-hour Zoom sessions over three months, sharing Java fundamentals from scratch through a slide-based curriculum I put together, and learning a lot about explaining ideas simply along the way
- Worked through a final project with the group using queue-based control structures, and was glad that several came away wanting to keep learning CS on their own

PROJECTS

Square Analytics Platform — TypeScript, Rust

Deployed · repo

- Built and deployed a **production full-stack analytics application** adopted by Walley's 9-person team to visualize transaction data, surface errors, and optimize inventory for a store processing **\$45,922** across **8,654 transactions**
- Implemented **high-performance Rust data pipelines** paired with a TypeScript frontend delivering real-time reporting and actionable insights to non-technical stakeholders
- Replaced a fully manual reconciliation process, saving significant administrative time weekly and **improving financial accuracy across the store**

Compilearn — AI & CS Education Platform — React, Vite, Tailwind, Tauri, Supabase, Groq API

Open Source · repo

- Built and launched a **free, open-source, browser-based platform** that teaches programming through **four hands-on tracks** — AI Engineering (Python), AP CSP, AP Computer Science A (Java), and Competitive Programming (C++) — born directly from 3 months of tutoring students who lacked access to quality instruction
- Designed an interactive **AI playground** where learners defend a model against adversarial prompts, alongside an **AI tutor**, both powered by the **Groq API** with per-user rate limiting and response caching to keep the entire platform free to use
- Built the frontend in **React, Vite, and Tailwind** with a **Supabase** backend and a cross-platform **Tauri desktop build**, and coordinated a peer beta-tester group refining UI/UX and curriculum against real learner gaps

Event Scheduler — Go, Python, Protocol Buffers

Open Source · repo

- Built an open-source scheduling engine for mock DECA conferences that converts registration spreadsheets into a complete schedule, **automatically assigning students to events, matching judges by expertise, allocating rooms, and slotting exams**
- Implemented a bottom-up constraint solver that orders judges from least to most flexible and groups largest-first, then packs assignments into time slots while **preventing overlapping and back-to-back events** for students in the same group
- Built parallel **Go and Python** implementations (~3,000 lines) over a shared **Protocol Buffers** schema, with CSV import, xlsx output, and unit tests, validated on a real run of **279 students and 43 judges**

DECA Form Checker — Python, Tesseract OCR, PDF parsing

Deployed · repo

- Built a tool that automatically verifies the **~600 signed registration PDFs** the chapter collects each year (three forms per student), sorting them into correct / incorrect / needs-review and replacing **10+ hours** of manual checking each cycle
- Detects whether each required field is signed and dated with two methods — counting vector path segments in text-layer PDFs and running **OCR** over the signature region for scans and photos — with every measurement scaled to the form's layout so phone photos and full-page scans both work
- Designed it to **fail safe** (marks a form correct only on positive evidence, since an unsigned medical release is worse than a false rejection), output per-form reasons and drafted resubmit emails, and run fully offline; validated across **13 labeled sample types**

Roommate Matcher — Python, pandas, C++

Deployed · repo

- Wrote a **C++** error detector for a DECA-conference room-assignment system that cross-checks every submitted registration form against the official **189-student** roster, flagging misspelled names, wrong grades, and whitespace/encoding issues before any matching runs
- Its output drives an auto-fix and reporting step that exports a cleaned roster and an Excel error report, cutting roughly **4 days of manual data-cleaning** per conference
- Debugged and repaired the core **Python/pandas** matcher, a pairing-of-pairs algorithm honoring mutual roommate requests under a **grade-adjacency constraint**, to ship the working build that sorts students into gender-separated rooms of four

HomeFit — Affordable Housing Locator — Rust, Tauri, React, TypeScript

Open Source · repo

- Built and deployed a **free, open-source desktop application** that pulls live HUD/government housing data, letting low-income users filter subsidized listings by location and income eligibility and view results on an interactive map, a solo project with a high-performance Rust backend and React/TypeScript frontend via Tauri
- Motivated by the belief that **navigating affordable housing should not require resources or technical literacy**, built to give any user, regardless of background, a free tool that centralizes data otherwise scattered and hard to act on

COMPETITIVE PROGRAMMING

- **Codeforces** — Current rating **1752** (Division 2), competing in rated contests and applying advanced algorithmic techniques under timed conditions (profile)
- **X-Camp Academy & USACO** — Enrolled in X-Camp's advanced algorithmic coursework to accelerate Codeforces rating growth and build competitive-programming depth; preparing for September and December 2026 USACO division assessments

COMPETITION RESULTS

- **Qualified for ICDC** — selected to compete at the **international level** in School-Based Enterprise, the highest tier of DECA competition
- **Silicon Valley Career Development Conference** — Top 10, against hundreds of regional competitors in business and entrepreneurship events
- **1st Place** — School-Wide Cultural Club Competition (Chinese Student Union, team of 5)

ACTIVITIES

Speech and Debate1 Year

- Synthesized complex case files and used Champion Briefs to construct **policy and argumentation strategies** for competitive regional tournaments
- Collaborated with debate partners on mock cross-examinations, sharing constructive critiques to elevate the team's competitive performance

Robotics (Roboquest)1 Year

- Executed hands-on **hardware-software integration**, applying engineering design processes to program and optimize autonomous mechanisms for competitive robotics
- Worked within a multidisciplinary engineering squad, bridging hardware assemblers and software programmers to ensure cohesive robot functionality

Chinese Student Union — Member3 Years · 1st Place (11th Grade)

Interests: Boxing (3 years), Guitar (4 years), Video Games (3 years)

- Contributed to a team of 5 designing and building a large-scale poster board display styled after a traditional Chinese dragon for the school's annual cultural club showcase
- Display earned **1st place** in a judged competition against 10+ cultural clubs, evaluated by school administrators before an audience of 100–150 students and parents

SKILLS

Programming	C++ (self-taught), TypeScript, Rust, Python, React, Next.js, Vite, Tailwind, Tauri, Supabase, competitive programming
Technical / CS	Data structures, algorithmic problem solving, full-stack development, data analytics, OCR/PDF parsing, performance engineering, open-source development
Business & Ops	Operations management, POS analytics (Square), inventory systems, revenue optimization, data-driven decision making
Teaching & Leadership	Curriculum design, technical mentorship, beginner-focused pedagogy, cross-functional leadership, agile methodologies, team orchestration
Languages	English (Fluent), Chinese (Proficient — translation and written correspondence)